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A Survey on Image Splice Forgery Detection and Localization Techniques

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***Abstract*—In this research, image splice detection and localization are explored. A spliced image is a digital image that has been copied and recreated on another image. The cropped image may undergo resizing, rotation, blurring, etc., before being forged. With the advancement of many photo-shop tools and software, image manipulation, and splicing have become a significant issue of concern. It is being misused nowadays in many fields like changing will, court orders, official documents, photo morphing, etc. Excellent forgeries are so deceptive that they go undetected by the naked eye. Traditional picture tamper detection algorithms many times fail to detect innovative image splicing. In the past few years, solutions based on deep learning have been proposed for tampered image detection and are found to be more efficient than other techniques. In this research paper, we review relevant research articles on image splicing, both detection, and localization, for the period 2003 to 2022.**

***Index Terms*—Image splice, forgery detection, splice localization, splice detection.**

I. Introduction

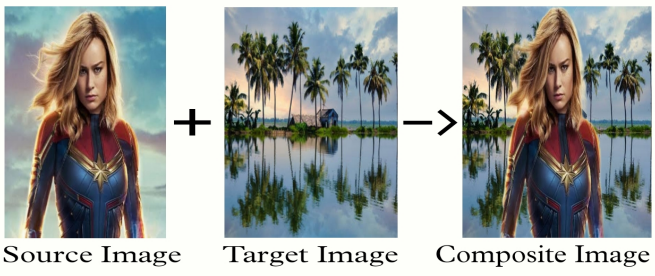


Fig. 1. Image Splice Forgery

Nowadays, with the advancement in technology and editing tools, image forgery has become simpler to create while detection has become more complex. Taking advantage of this, many image forgery based crimes such as morphing will change in court etc., are taking place widely and hence it has become a necessity to detect such manipulated images.
Figure 1 above depicts an example of a spliced image. Here, the source and target images create composite or forged images. As we can see here, the girl from the first image is fused onto the second one among the trees to create the spliced image. This scenario of creating forged images using two or more image sources is a clear depiction of image splicing.
In the recent past, image splicing, both detection and localisation, has flourished especially with advent of deep neural networks. In this research work, we summarise important research and popular data sets used in image splicing during the period 2003 to 2022. We believe that this paper is going to give a good starting point for those who are interested in the domain of image splicing. There are several methods which are currently being used for the same, such as Local Binary Pattern and Discrete Cosine Transform CNN based feature extraction, by using noise difference from camera capturing and many other techniques [22], that are being discussed in this paper.

II. Image Forgery Detection Techniques

There are several methods developed for detecting forgery in digitally scanned documents. The active and passive approaches are the most common classifications (Fig. 2).
The active approaches mainly focus on data concealment techniques, such as digital watermarking or copyrighting, in which previous knowledge is required as critical and necessary to the process. At the same time, passive methods do not need any knowledge of the original image. In passive blind approaches, the detector only has access to the end product. This approach only uses the given input image to extract information such as texture, features, color, etc. Information regarding the original image is not required, hence known as a blind approach.

III. Classification of image forgery

There are three types of image manipulation techniques that come under the passive approach and are:

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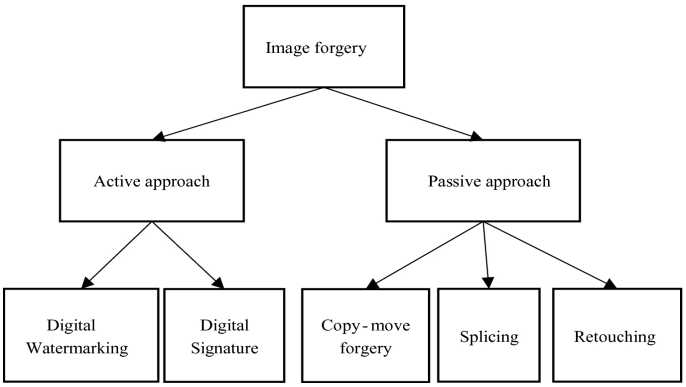


Fig. 2. Image Forgery classification Flowchart

A. Copy-Move Forgery:

One among the most well-known types of image forgery is the copy-move forgery. Here a part of an image is copied and pasted onto another part of that image itself. An example is shown in fig. 3, where the duck from the same image was copied and pasted behind it. Many techniques have been found for the efficient detection of copy-move image forgery [24].

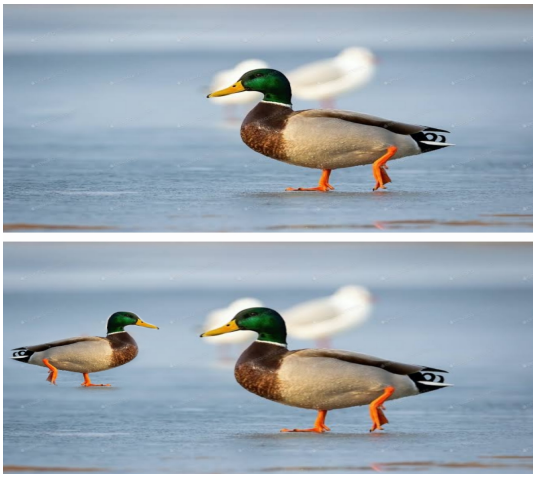


Fig. 3. Copy Move forged image

B. Image Retouching:

Retouching includes enhancing or reducing certain features of an image by removing all flaws from a photograph, which commonly entails colour and tone correction, the removal of blemishes and under-eye circles, deleting parts of an image, as well as changes to brightness, contrast, and saturation. In Fig. 4, we can see how one of the coconut trees has been removed and saturation has been increased compared to the adjacent image.

C. Image splicing:

It is a forgery technique in which the forger copies or cuts a part of a picture and pastes it onto another image. Here two image sources are hence required to create the forged image.



Fig. 4. Example of image retouching

Several illegal activities are being carried out by misusing this technology, such as changing of will, image morphing, misusing, etc. Fig. 1 shows how splicing is done on digital images. Several techniques have been invented for tackling down the same.

IV. Image Splicing

A. Data sets used for image splice detection

There are many public data sets for image splice forgery detection and localisation studies. A summary of the same can be found in Table 1, which depicts the most commonly used image splice data sets used by researchers worldwide.

TABLE I
MOST COMMONLY USED IMAGE SPLICING DATA SETS

Data Set	Size	No. of images	Format
CASIA v1.0 [27]	384 x 256	1721	JPG
CASIA v2.0 [27]	240×160 to 900×600	12,614	JPG, TIFF, BMP
Columbia [8]	128 x 128	1845	JPEG
DSO-1 [18]	2,048 x 1,536	200	PNG
IFS-TC [26]	1024×768 to 2848×2144	2200	PNG

- **CASIA v1.0 [27]:** The actual set has 800 photos. The modified set contains 921 pictures. Before pasting, cropped picture region(s) are treated with scaling, cropping, rotating, etc., to create spliced images. After creating a spliced picture, no post-processing techniques (such as blurring) is used in this data set.
- **CASIA v2.0 [27]:** There are 12,323 color photos and two image subsets in this collection (authentic and tampered). There are 7,200 legitimate photographs in the authentic set and 5,123 altered images in the tampered set. In addition to splicing, database V2.0 introduces blurring while modifying the altered image set [25]. In contrast to CASIA V1.0, the photos in CASIA V2.0 are of different sizes. CASIA V2.0, unlike V1.0, includes several uncompressed picture examples (BMP and TIFF) and JPEG images with various Q factors.

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- **Columbia Image Splicing Detection Evaluation Data set [8]:** It contains 933 genuine and 912 spliced 128 x 128-pixel picture chunks. The natural and spliced photos have almost the same amount of pixels. Crop-and-paste along object boundaries vs. horizontal (or vertical) strips have been used to replicate the process of producing spliced pictures. Picture objects and strips might come from the same or two different image sources. Objects might be of the same or distinct sorts, textured or smooth and are spliced together.
- **DSO-1 [18]:** There are 100 genuine photos and 100 forged photographs in this collection. Both interior and exterior photos are included in the collection. The fictitious photos are created by inserting one or more people into the original photograph. Various operations, such as color and lighting changes, are carried out to create believable fabricated pictures.
- **IFS-TC [26]:** This data set was first utilized in an IFS-TC-sponsored international competition. It contains 1150 forgeries and 1050 genuine photos.

B. Image Splice Forgery Detection Techniques

Most traditional splice forgery detection techniques are not efficient enough to detect image tampering. However, modern deep learning techniques that use CNN-induced methods are more efficient and effective [18]. Due to the complexity in detection spliced images, several researchers came up with different approaches [6]. Some only detect spliced forgery regions, while others can detect and localize spliced regions. Some techniques even detect blurred images after splicing [19].

One standard method to detect image splice forgery is using LBP and DCT approach [1]. Here, the local binary pattern corresponding to an image is initially computed using its YCbCr chrominance, and from that, it is transformed to its frequency domain by applying DCT. Then it is sent for further splice detection. Another standard method referred to was that of detecting spliced images via camera noise detection [2]. Another technique referred to was that of reliability fusion maps, which is efficient for localization of even tiny patches of image pixels [3]. ELA (Error Level Analysis) is yet another effective method for splice detection [4] [5], which, when combined with other localization algorithms [23], can provide more effective outputs. In the recent past, deep convolution neural network has set the best results in all benchmark data sets. CNN based papers were the most cited due to it's effective feature extraction capability and classification [21]. In the following subsection, we review starting from feature based techniques to the more recent deep neural network based methods.

1) Feature Based Techniques: Feature extraction is one of the early method used in detecting image splicing. Over the years, many techniques have been proposed for image splice detection using feature-based techniques. One of the early features used in image splice detection is the Local Binary Pattern (LBP) [1], Error Level Analysis (ELA) [4] [5], Discrete Cosine Transform (DCT), Markov Transform model, etc.

Feature-based techniques are used to enhance the features of an image; feature extraction becomes more effortless, and so does the classification. This stage is known as the pre-processing stage. This pre-processing stage is optional in most cases, as shown in the Fig. 5. However, effective use of this stage can give better results. Some of them are given in table II.

In [1], an image splice detection method using LBP along with DCT is proposed. The LBP code of each block is found after the chromatic image component is divided into overlapping blocks, and it is then transformed into its DCT domain. Later the SVM classifier is being used here for classification. We can observe that using this technique, the authors have reached an accuracy of 97.5 on the CASIA V2.0 dataset.

In [4], a technique combining both Error Level Analysis and Convolution Neural Network is proposed. First, the data set is divided into forged and original images, after which VGG 16 is used to train because VGG is excellent for training with small data sets. Although an excellent deep learning model is used for splice detection, we can observe from the results that architecture that uses a combination of two or more pre-processing stages is getting better results.

In [5], they have proposed a technique based on ELA and LBP for detecting spliced images. Instead of one, two feature extractors are used on the input image, i.e., ELA and LBP. Then, these features are fused and fed into the classifier, and these features are classified into authentic and tampered images. Finally, they accurately locate the tampered area through the improved ELA method. The authors suggest at the end that if this combination of ELA and LBP is implemented on CNN, there are higher chances of getting even better accuracy.

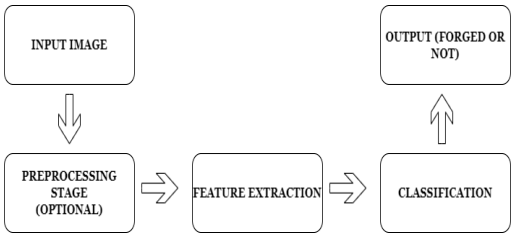


Fig. 5. Commonly used image forgery detection method

2) CNN based methods: Techniques based on convolution neural networks are rapidly improving in terms of performance [16]. Over the years of comparison of results, we can observe that the results are improving rapidly with the advancement of technology [28]. So is the case with Convolution Neural Networks. Initially, CNN was used as a substitute for classifiers such as SVM, but nowadays, CNN is being effectively used for localization of the spliced area accurately [20] [28]. It has been achieved by inducing more complex models such as

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TABLE II
FEATURE BASED TECHNIQUES COMPARISON

Feature based techniques comparison				
Reference Paper	Year	Method	Data set	Perfomance
Alahmadi, A. A. et al. [1]	2013	LBP,DCT ,SVM classifier	CASIA 2	Accuracy-97.5
Sudiatmika, I. B. K. et al. [4]	2019	ELA,CNN	CASIA 2	Accuracy-88.46
Zhang, Y.-J. et al. [5]	2021	ELA, LBP ,Bagged Trees classifier	MICC-2000	Accuracy-97.30

RRU-Net, ManTra-Net, etc. Strategies of specific CNN-based splice detection models that have achieved high accuracy are described in this section.

In [6], an end-to-end network model is proposed for both detection as well as localization of spliced images without any other pre or postprocessing. ManTra-Net is a complete convolution network that can handle photos of any size and various forgery kinds, including splicing, copy-move, removal, enhancement, and even unknown forging forms. ManTraNet comprises two sub-networks: Image manipulation trace feature extractor, which is used to detect image manipulations, and local anomaly detection network, which is used to train a decision function mapping from a local feature's reference to its forgery label.

Because of their limited computational speed and memory availability, existing deep learning models can only accept tiny pictures as input, necessitating pre-resizing. Resizing has little effect on discriminative characteristics in high-level vision difficulties; therefore, this is not a concern. In image forensics, on the other hand, scaling tends to remove valuable high-frequency features, which has a significant influence on speed. Patch-wise processing can prevent resizing at the expense of whole-image analysis. For accurate picture forensics, [14] was proposed. The performance of a constrained R-CNN architecture is shown in the table below, executes manipulation techniques classification and changed region segmentation at the same time. The learnable manipulation feature extractor starts by taking data and extracting a unified feature representation. Second, for future manipulation classification and coarse localization, the attention region proposal network successfully discriminates changed areas. By merging low-level and high-level information, the skip structure refines the global manipulation features. Finally, localization data instructs the model to acquire more acceptable local characteristics and segment the tampered area.

In [7], the model consists of A CNN-based approach for picture forgery detection. It enables the processing of integrated data obtained from the whole image at full resolution . As a consequence, the authors propose a three-part framework: patch-wise feature extraction, image-wise feature aggregation, and global judgement. Although deep learning models perform well in detecting picture alteration, most of them suffer from a lack of universality of specified characteristics. In the meantime, they are solely concerned with manipulation localization and do not consider manipulation classification.

TABLE III
CONVOLUTION NEURAL NETWORK BASED TECHNIQUES COMPARISON

Convolution Neural Network based techniques				
Reference Paper	Year	Method	Data set	Perfomance
Wu, Y. et al. [6]	2019	ManTra-Net	COVERAGE, CASIA, NIST, Columbia	Accuracy- 85
Yang, C. et al. [14]	2019	Constrained R-CNN	NIST16, Columbia, COVER, CASIA [15]	F1 Score - 0.927, AUC - 0.992
Marra, F. et al. [7]	2020	CNN	Vision/UCID, Dresden/FAU, DSO-1 ,Korus, NC2017, MFC2018	AUC - 0.93
Rao Y. et al. [29]	2020	CNN	CASIA v2.0, DSO-1	Accuracy-97.50
Pandey A. et al. [28]	2022	CNN, Image transformation	CASIA v2.0	Accuracy- 93

3) Forgeries based on inconsistency in noise levels of image: Image splicing can also be detected using the noise inconsistencies within the input image. This technique is based on the variation in noise consistencies of different images depending upon the place, time, and model with which it has been taken. The table shows that detection accuracy has increased over the years with the efficient utilization of noise in images for detection of forgery.

In [9], the image is split into non-overlapping blocks, which are then clustered to create fresh and modified chunks. To get at the final result, the found suspicious zones are subdivided again to refine noise estimation and then classified. The author segmented the image into sections with considerably different noise variances to enhance the results even further to estimate local noise variances. A simple k-means clustering technique enhanced the outcome, followed by post-processing processes on the foundation areas. In [10] a method that is automated Spliced forgery detection in raw photos has been suggested. They looked at the relative consistency of noise parameters. To identify the spliced pictures, image discrepancies from quad-tree decomposition. In [8], a method based on the premise that the level of noise in the original and the image to be spliced varies. The picture is first divided into superpixels and then extracted into ordinary forms. Three color channel differences are recovered from each superpixel, and the residual noise level is evaluated. Finally, the superpixels are classified as tampered or original by clustering them into two classes using the Farthest Distributed Centroids Clustering algorithm.

4) Statistical properties inherent in the source image: Image splice detection can also be found using the statistical properties inherent in the source image. It is entirely dependent

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TABLE IV
NOISE BASED TECHNIQUES COMPARISON

Noise based techniques				
Reference Paper	Year	Method	Data set	Perfomance
Pan, X. et al. [9]	2012	DCT, Inconsistency in image noise levels	KODAK	Block Detection Accuracy - 70
Julliand, T. et al. [10]	2015	relative consistency of noise parameters	own data (real data)	Accuracy - 84
Chandra Sekhar, P. et al. [8]	2019	Colour channel extraction , noise level estimation ,FDCC for classification	Columbia	F1 score-71.9

on the time when the picture has been taken; like camera model- is it same or different, camera response function, etc. In [2] a new technique was proposed since depending on image splice detection based on watermarking alone was not successful. Here doctored images are found using the camera response normality function. If the response functions are incompatible with each other, the picture is doctored. A support vector machine is used to classify the normalcy of the response functions.

In [11], in contrast to previous approaches, this technique is based on a human visual system (HVS) model in where the feature extraction process is guided by visual saliency and fixation. The authors propose using visual saliency and fixation to drive the feature extraction technique in order to distinguish a spliced image. A noteworthy component of this idea is the great link among the splicing borders and the first few fixation sites detected by a visual attention model using edge sharpness, which contributes in obtaining improved accuracy.

TABLE V
SOURCE IMAGE PROPERTIES BASED TECHNIQUES COMPARISON

Source image properties based techniques comparison				
Reference Paper	Year	Method	Data set	Perfomance
Lin, Z. et al. [2]	2005	Camera response function, SVM	DoRF	SVM range : 0.53 to 1.77
Qu, Z. et al. [11]	2009	HVS	Columbia	Accuracy- 96.33

5) Illumination color Estimation: Detecting image splicing using illumination color estimation is yet another technique commonly used. This technique is implemented by using the physical environment of the image taken. Image splicing can be easily identified using Illumination color Estimation [18], where there will be changes in the illumination of the background, say sunset or sunrise, and color changes, if any. Hence if the images are taken from the same lighting condition and colors, the detection accuracy may reduce. In this type of method, too, accuracy increases as color detection efficiency increases, such as pixel and edge-based illuminance color estimation. In [13] we can see that the chrominance and luminance of the input image is determined, and illumination is calculated from luminance using the Illumination-Reflectance model. Finally, the feature vector is found using LBP for classification.

In [18], we can find a new approach was induced for detecting photographic splicing that combines the excellent representation capability of Illuminant Maps and convolution neural networks as a way of learning the essential signals of a counterfeit straight from accessible training data. The authors have obtained an accuracy of 96 for the same. In [12] the spliced pictures are discovered on image areas utilising pixel and edge-based illuminant colour estimation. The shape and colour characteristics are retrieved from these illuminant estimators, then fed into SVM for classification.

TABLE VI
ILLUMINATION COLOUR ESTIMATION BASED TECHNIQUES COMPARISON

Illumination colour estimation based techniques comparison				
Reference Paper	Year	Method	Data set	Perfomance
Youseph. et al. [13]	2015	illuminant estimators,SVM	CASIA 2	Accuracy- 74
Pomari, T., et al. [18]	2018	Illuminant Maps, CNN	DSO	Accuracy- 96
Niyishaka. et al. [12]	2020	Luminance, LBP,Linear Discriminant Analysis (Classifier)	CASIA v2.0	Accuracy- 94.59

V. CONCLUSION

Although there were several techniques implemented for the successful detection of image splicing, there was a lacking of an effective review paper stating and comparing the best techniques in use. Hence this paper was bought up to fill up that gap in the research field of image splice detection as well as localization. In this paper some of the important research works in image splice detection during the period of 2003 to 2022 were reviewed. Several techniques by which image splice detection as well as localization are found were also discussed and compared. After going through several literature reviews, it was noted that there are several traditional methods, such as watermarking, camera response, illuminance based techniques, lighting based methods, RGB/YCbCr color component, etc., used in the detection of spliced images. However, we can observe that techniques that use CNN within it surpasses all other techniques in terms of cost and accuracy. So a CNN-induced method, when used successfully by imposing onto other traditional algorithms, can result in an excellent image splice detection and localization, which shall be of great help in the future.

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